

COSE471: Data Science
Term Project

Department of Computer Science and Engineering
Korea University
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June 5, 2026

Seating Plan by Team (check your team in LMS)

Attendance code:

Front

1	Team 1
2	
3	Team 2
4	
5	Team 3
6	
7	Team 4
8	
9	Team 5
10	

1	
2	Team 6
3	
4	Team 7
5	
6	Team 8
7	
8	Team 9
9	

1	Team 10
2	
3	Team 11
4	
5	Team 12
6	
7	Team 13
8	
9	Team 14
10	

Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1		<i>Latex, Preprocessing/Pattern mining</i>
5	Apr 03	Clustering		
6	Apr 10	Outlier Detection & Invited Talk		<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW (1&2 merged)	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification		
9	May 01	Classification		
10	May 08	Tutorial 3	●	<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project	
13	May 29	Project discussion		
14	Jun 05	Project discussion		
15	Jun 12	Final presentation		
16	Jun 19	Final report (No class)	●	

Term Project

❑ REAL-WORLD in-class discussion datasets

- 5 topics, 9 questions, 294 responses, 157k characters from ALL of YOU (69 students)!

❑ Goal

- Suggest a unique, interesting, fun data science research question to explore using the dataset
 - e.g., “*can we map the landscape of student responses and identify similar groups?*”
- Answer the question by applying the core concepts and methods covered in this course
 - Mandatory: data understanding and preprocessing
 - Choose at least two: pattern mining, clustering, outlier detection, classification
 - Optional: advanced tools/techniques practiced with in tutorials
 - Note: Any other techniques outside the scope of this course should come with sufficient explanation and justification

❑ Deliverables

- Presentation on June 12 (offline for selected teams, video recording upload by 9:00 AM for other teams)
- Report by June 20 11:59 PM (up to 10 pages in PDF compiled using Latex) – *free template*

Term Project

❑ Recommended structure for the final report

- Introduction (i.e., background, motivation, problem setting, goal, challenges)
- Methodology (i.e., data understanding/preprocessing, pattern mining, clustering, outlier detection, classification)
- Results (i.e., quantitative/qualitative results and visualization)
- Discussion and conclusion (i.e., findings, insight, feedback, limitations)
- Roles and responsibilities of each member

❑ Criteria for reference

- Novelty (i.e., how unique and interesting your problem, results, and findings are)
- Technical quality (i.e., how well you apply and adapt the concepts/methods covered in class to your specific problem)
- Presentation (i.e., how clearly you explain your project, both verbally in your offline presentation and in writing in your report)
- Reproducibility (i.e., how thoroughly you document your approach and procedures, providing sufficient justification and detail)

❑ ***Note: if you need a commercial LLM API (e.g., GPT-5 mini) for project, please let me know.***

Project Discussion (May 29)

□ Today's mission for all teams

1. Say hi to your team members and share contact points
2. Set your team nickname
3. Brainstorm questions your team want to answer using the dataset
4. Plan each members' role and timeline; it can be tentative
5. Submit the summary of your discussion (*used for the 2nd attendance check*)
 - Must include: team nickname, question candidates, role/timeline plan
 - *Once completed, you can leave anytime!*

+) Bonus mission only for random teams

- Take a photo of your *offline* team meeting *outside of the regular class hours (not today!)*
 - For those who are present in the photo (please list their names), 1% extra point will be given
 - If all members join the meeting at off-campus places (please specify the location), you will get additional 1% extra point
- Submit one photo per team to susik@korea.ac.kr by June 12

Project Discussion (June 5)

□ Today's mission for all teams

1. Briefly review the summary from last class and confirm everyone is clear on team's goal and their roles.
2. Finalize the research question the team will answer in the term project (this can be changed, if necessary)
3. Identify core challenges in addressing the research question, such as
 - *Does the dataset include sufficient information to answer the question?*
 - *Do you have plausible intuition or ground-truth answers that can be used to judge the quality of results?*
4. Brainstorm how to overcome or work around the challenges, such as
 - *How can we adapt and exploit the concepts, methodologies, or techniques introduced in the lectures and tutorials?*
 - *Can we use any external data or tools to supplement the given dataset?*
5. Submit the summary of your discussion (*used for the 2nd attendance check*)
 - Must include: team nickname, a research question of your team, core challenges for the research question.

Once completed, you can leave anytime!